Congressional District Characteristics Associated with House Turnout and Voter Preference

A Multivariate Regression Analysis of the 2024 U.S. House Elections

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November 8, 2026

1 Introduction[1]

Politics has become far more polarized in this century as demonstrated by the increased frequency in changing party majorities in the House: changing four times in the last 25 years compared to only once in the previous 25 years (Wikipedia contributors 2026).

In light of this increased political volatility, this paper evaluates the congressional district demographic characteristics and their potential association with voting outcomes. The number of votes for a party's candidate can be viewed as a function of two outcomes: the proportion of eligible voters who turn out to vote, and the share of votes cast for each party. To investigate potential drivers of these outcomes, the following research questions are investigated:

- Which district characteristics are associated with turnout and two-party voter share, taken together?
- Do thse characteristics impact voter turnout and party share to the same degree?

Political science research has investigated these questions separately, using univariate regression models, ignoring any correlation between turnout and vote share. To evaluate whether modeling the outcomes jointly provides additional insights, a multivariate regression model is used to estimate the correlation between the two outcomes' errors and to test the joint effect of each predictor on both outcomes, comparing results of the univariate models.

The data cover the 2024 House elections which elected the 119th Congress. Election returns were obtained from the MIT Election Data + Science Lab (MIT Election Data and Science Lab 2017), demographic and economic characteristics of each district were obtained from the Census Bureau's American Community Survey (ACS) 5-year estimates (2020–2024) (U.S. Census Bureau 2026), and land area from the Census Bureau's 2024 Gazetteer file (U.S. Census Bureau 2024).

The results suggest that turnout and voter share (baseline defined as Democratic share) respond differently to district characteristics. The predictors explain most of the variation in turnout (70.9%) and Democratic share (82.0%), and the joint test that no predictor is associated with either outcome is rejected using Pillai's trace ($p < 0.001$). Nine of the twelve predictors are associated with the two outcomes by significantly different amounts. The residual correlation between the outcome is near zero so that individual and joint models reach the same conclusion for 11 of 12 predictors (exception

[1]All R scripts, data and report files are available at https://github.com/janblackbu69/Math257.

Table 1: Descriptive statistics for the outcome and predictor variables (n = 396 contested districts)

Variable	Mean	SD	Min	Median	Max
Turnout (%)	62.40	8.70	37.20	63.20	81.10
Democratic two-party share (%)	49.90	15.70	19.60	49.80	88.70
Black share (%)	12.30	13.10	0.30	7.90	65.40
Hispanic share (%)	14.90	16.00	1.00	8.70	86.80
Asian share (%)	5.10	6.80	0.30	2.60	51.40
Bachelor's degree or higher (%)	35.60	11.00	10.20	33.90	80.30
Median household income ($10,000s)	8.44	2.19	4.45	7.90	18.22
Income inequality (Gini index x 100)	46.20	2.90	37.20	45.90	58.00
Age 18-29 (%)	20.20	2.90	12.00	19.90	30.80
Age 65 and over (%)	22.10	4.00	11.80	22.10	41.30
Female share (%)	51.20	1.20	47.00	51.20	55.10
Population density (log10)	2.66	0.81	0.11	2.58	4.88

is female share). The joint model provides greater insight into how predictor variables have different impacts on turnout and voter share.

The remainder of this paper describes the data, summarizes the methods used to model and evaluate the outcomes, presents the results, and concludes with a discussion of their implications and potential follow-on research.

2 Data

The unit of analysis is the congressional district, using the 119th Congress districts from the November 2024 election, in which Democrats gained a net two seats to reach 215 while Republicans retained control with 220 seats (Wikipedia contributors 2026). The two outcome variables are turnout, calculated as the number of votes cast in the House race divided by the citizen voting-age population (CVAP), and Democratic share, calculated as Democratic votes divided by the sum of Democratic and Republican votes.

The model uses 396 of the 435 districts, excluding 39 uncontested districts in which two-party share is undefined since these districts were uncontested. The predictor variables describe the racial, educational, income, age and gender composition of each district, its population density and the party of the incumbent (see Appendix A). The non-Hispanic White share is omitted out since it's nearly a linear combination of the other race shares. Three other economic measures (low-income share, high-income share and poverty rate) are left out due to collinearity with median income. Incumbency comes from 2022 results with 170 Democratic incumbents, 177 Republican incumbents and 49 open seats (baseline). Manfy of the excluded 39 districts are mostly concentrated in Massachusetts (7 districts), Texas (8 districts) and California (2 districts). No districts were dropped for missing Census data.

Descriptive statistics are shown in Table 1. The correlation between turnout and Democratic share is -0.19, indicating that Democratic districts tend to have lower turnout. The female share varies little across districts (SD = 1.2 percentage points), which gives the model little information with which to detect its effect.

3 Methods

3.1 Univariate models

This approach fits a separate multiple regression for each outcome k, where $k = T$ for turnout and $k = D$ for Democratic share:

$$\mathbf{y}_k = \mathbf{X}\beta_k + \epsilon_k, \qquad \epsilon_k \sim N_n(\mathbf{0}, \sigma_k^2 \mathbf{I}_n), \qquad k \in \{T, D\} \tag{1}$$

- $\mathbf{y}_k$ is a list of one outcome's values, one number per district. For turnout ($k = T$), it holds every district's turnout; for Democratic share ($k = D$), every district's Democratic share.
- $\mathbf{X}$ is the table of predictor values, with one row per district and 13 columns. The first column is all 1s to model estimate an intercept (the baseline level of the outcome). The other 12 columns hold the predictors, such as Black share, education and density.
- β_k is the list of 13 coefficients for outcome k: the intercept plus one coefficient per predictor. Each coefficient says how much the outcome changes when that predictor goes up by one unit, holding the others fixed.
- ϵ_k is the part of each district's outcome the predictors don't explain. The model assumes these errors are independent from district to district, follow a normal distribution, and have the same variation (σ_k^2) across districts.

3.2 Multivariate model

The multivariate model regresses both outcomes on the same twelve predictors at once:

$$\mathbf{Y}_{n\times 2} = \mathbf{X}_{n\times 13}\,\mathbf{B}_{13\times 2} + \mathbf{E}, \qquad \varepsilon_i \sim N_2(\mathbf{0}, \Sigma), \qquad \Sigma = \begin{pmatrix} \sigma_T^2 & \sigma_{TD} \\ \sigma_{TD} & \sigma_D^2 \end{pmatrix} \tag{2}$$

- $\mathbf{Y}$ holds one district's turnout and Democratic share
- ε_i is the ith row of $\mathbf{E}$.
- $\mathbf{B}$ is the coefficient matrix and has one column per outcome. Because both equations use the same predictors, the estimates are identical to those from the two separate OLS regressions.
- ϵ_k is the errors and includes a covariance matrix Σ, which allows a district's turnout and voter share to be correlated.

Σ is a 2×2 table describing the errors, the part of each outcome the predictors don't explain:

- σ_T^2 (top left) is the variance of the turnout errors: how far districts' actual turnout typically lands from what the model predicts.
- σ_D^2 (bottom right) is the same thing for Democratic share.
- σ_{TD} (both off-diagonal elements) is the covariance between the two errors. It describes whether a district that beats its predicted turnout also tends to beat, or fall short of, its predicted Democratic share.

3.3 Joint significance tests

For each predictor j, the joint test asks whether it is associated with either outcome, testing $H_0 : (\beta_{jT}, \beta_{jD}) = (0, 0)$ with Pillai's trace (Johnson and Wichern 2007):

$$V = \text{tr}[\mathbf{S}_H(\mathbf{S}_H + \mathbf{S}_E)^{-1}] \tag{3}$$

For each predictor, the test is

$$H_0 : \beta^*_{jT} - \beta^*_{jD} = 0, \tag{4}$$

using the response contrast $\mathbf{c} = (1, -1)^\top$. Rejection means the predictor's effect differs between the outcomes.

Because turnout and Democratic share have different spreads, both are standardized to mean 0 and SD 1, so each coefficient (β^*) shows how many standard deviations an outcome moves per one-unit change in the predictor, putting the two outcomes on the same scale. The test then asks whether a predictor's two standardized coefficients are equal, which is the same as regressing each district's difference between its two standardized outcomes ($z_T - z_D$) on all the predictors and checking whether that predictor's coefficient is zero; the F statistic is simply that coefficient's t statistic squared. The test requires the covariance between the two coefficient estimates, which two separate regressions don't provide, so it's only possible when both outcomes are modeled together.

3.4 Assumptions and diagnostics

Multivariate regression requires that the error vectors ε_i be independent across districts, that they share a constant covariance matrix Σ, and that they be bivariate normal. The two errors within a district may be correlated as captured by Σ. The large sample size ($n = 396$) means moderate departures from normality should still give reliable test statistics.

Diagnostic checks: Plots of residuals against fitted values check that the relationships are linear and that the errors have a similar spread everywhere; a curve or funnel shape signals violations in the constant variation assumption. Normal Q-Q plots of the standardized residuals check whether each outcome's errors are normal, which they are if the points fall along a straight line. To check whether the two errors are jointly normal, each district's pair of residuals is reduced to a single number, the squared Mahalanobis distance, which measures how unusual the pair is after accounting for each outcome's spread and the correlation between them; if the errors are bivariate normal, these distances follow a chi-square distribution with 2 degrees of freedom, so plotting them against that distribution's quantiles should also follow a straight line.

Multivariate vs. Univariate: How much the multivariate model adds depends on the residual correlation between the outcomes. If that correlation is weak, the joint model should give nearly the same answers as the separate models. The joint model helps most when the correlation runs against the pattern of a predictor's effects, for example, when a predictor raises both outcomes but their errors are negatively correlated. It can lose power when a predictor affects only one outcome, because it spreads its attention across both.

All analysis was done in R (R Core Team 2025) with the tidyverse (Wickham et al. 2019), car (Fox and Weisberg 2019), sandwich (Zeileis et al. 2020), lmtest (Zeileis and Hothorn 2002) and broom (Robinson et al., n.d.) packages, and tables were formatted with kableExtra (Zhu, n.d.).

Table 2: Separate regression coefficients with t-test p-values, and joint Pillai tests for each predictor (classical standard errors, n = 396)

Predictor	Turnout b (SE)	Turnout p	Dem. share b (SE)	Dem. share p	Pillai	Joint p
Black share (%)	0.021 (0.033)	0.528	0.390 (0.047)	< 0.001	0.152	< 0.001
Hispanic share (%)	-0.065 (0.021)	0.003	0.140 (0.031)	< 0.001	0.076	< 0.001
Asian share (%)	-0.426 (0.050)	< 0.001	0.224 (0.071)	0.002	0.183	< 0.001
Bachelor's degree or higher (%)	0.524 (0.055)	< 0.001	0.401 (0.078)	< 0.001	0.228	< 0.001
Median household income ($10,000s)	0.264 (0.283)	0.352	-0.284 (0.403)	0.481	0.004	0.488
Income inequality (Gini index x 100)	-0.908 (0.126)	< 0.001	-0.121 (0.179)	0.500	0.120	< 0.001
Age 18-29 (%)	-0.745 (0.145)	< 0.001	0.839 (0.206)	< 0.001	0.106	< 0.001
Age 65 and over (%)	0.312 (0.117)	0.008	0.585 (0.166)	< 0.001	0.046	< 0.001
Female share (%)	-0.528 (0.378)	0.164	-1.149 (0.538)	0.033	0.016	0.046
Population density (log10)	-1.374 (0.500)	0.006	3.843 (0.712)	< 0.001	0.091	< 0.001
Democratic incumbent	-0.153 (0.814)	0.851	8.929 (1.159)	< 0.001	0.135	< 0.001
Republican incumbent	-0.320 (0.782)	0.683	-8.622 (1.114)	< 0.001	0.135	< 0.001

4 Results

4.1 Model fit and joint tests

Overall, the models fit the data well. The predictors explain 70.9% of the variation in turnout (adjusted $R^2 = 0.700$; $F(12, 383) = 77.9$, $p < 0.001$) and 82.0% in Democratic share (adjusted $R^2 = 0.814$; $F(12, 383) = 145.5$, $p < 0.001$). The joint test that all twelve predictors have no effect on either outcome is decisively rejected (Pillai's trace = 1.51; $F(24, 766) = 97.7$, $p < 0.001$).

Coefficients are expressed in percentage points of the outcome per one-unit change in the predictor. For example, a district with 10 more percentage points of college graduates has 5.2 points higher turnout and 4.0 points higher Democratic share, holding the other predictors constant. Table 2 summarizes the separate t-tests and the joint Pillai test for each predictor.

The joint and separate tests reach the same conclusion for 11 of the 12 predictors: ten are significant both ways but median income is insignificant. The one exception is the female share, which misses the Bonferroni cutoff in both separate tests ($p = 0.164$ and 0.033, against 0.025) but is jointly significant ($p = 0.046$), because its two negative effects together provide just enough evidence; given how little this variable varies and how close the p-value is to 0.05, it's a weak result. The two approaches agree so closely because the residual correlation is nearly zero ($r = 0.054$, $p = 0.29$), compared with the raw correlation of -0.19, meaning the predictors explain almost all of the relationship between turnout and Democratic share.

4.2 Cross-outcome tests

Having identified which predictors are jointly significant, the next hypothesis tested is whether each predictor is associated with turnout and voter Democratic share by the same amount (null hypothesis) or not. Table 3 summarizes the results on the standardized outcomes. Nine of the twelve predictors have significantly different effects on the two outcomes ($p < 0.001$) showing clear rejection of the null hypothesis, while the effects of median income, the 65-and-over share and the female share are not significantly different. Four predictors (Hispanic share, Asian share, the 18–29 share and population density) are significantly associated with both outcomes, but in opposite directions.

Table 3: Cross-outcome tests: effect of each predictor on standardized turnout and Democratic share (SD change per one-unit change in the predictor), with F tests of equal effects

Predictor	Turnout (SD)	Dem. share (SD)	F(1, 383)	p
Black share (%)	0.002	0.025	22.50	< 0.001
Hispanic share (%)	-0.007	0.009	28.60	< 0.001
Asian share (%)	-0.049	0.014	78.90	< 0.001
Bachelor's degree or higher (%)	0.061	0.026	19.90	< 0.001
Median household income ($10,000s)	0.030	-0.018	1.40	0.231
Income inequality (Gini index x 100)	-0.105	-0.008	29.10	< 0.001
Age 18-29 (%)	-0.086	0.054	45.30	< 0.001
Age 65 and over (%)	0.036	0.037	0.01	0.936
Female share (%)	-0.061	-0.073	0.05	0.819
Population density (log10)	-0.159	0.245	31.90	< 0.001
Democratic incumbent	-0.018	0.570	25.50	< 0.001
Republican incumbent	-0.037	-0.551	21.00	< 0.001

4.3 Model diagnostics

One limitation of the multivariate regression model is the strong set of assumptions it requires, but the analysis suggests the data reasonably meet them. Figure 1 (Appendix B) plots the residuals against the fitted values for each outcome. Neither panel shows curvature or a funnel shape. Fitted values for Democratic share fall into two clusters, below and above 50%, reflecting the shift of about 9 points associated with Democratic and Republican incumbents, but the spread of residuals is similar in both clusters.

Normality of the residuals supports the validity of the t-tests. Figure 2 (Appendix B) compares the standardized residuals with theoretical normal quantiles. Both outcomes follow the normal line closely, with slightly heavy upper tails, but given the large sample size, these departures are unlikely to affect the results. Figure 3 tests whether each district's pair of errors for turnout and Democratic share is bivariate normal by plotting the squared Mahalanobis distances against the quantiles of the $\chi^2(2)$ distribution. The points track the reference line closely, with a slight bulge between quantiles 5 and 7 and only a single clear outlier. The bivariate normality assumption appears reasonably met, and the Pillai tests are unlikely to be distorted by non-normality given the large sample.

5 Discussion

The primary finding of the analysis is that turnout and Democratic share respond differently to district characteristics. Democratic share is associated with race, density, age and incumbency, while turnout is associated with education, inequality, and the Asian and young-adult shares of the electorate. Only education strongly raises both. Several predictors pull the two outcomes in opposite directions. Denser, younger and more Asian districts vote more Democratic but have lower turnout. This helps explain the negative correlation between turnout and Democratic share (-0.19): it is a product of where Democratic-leaning groups live, not a direct link between the outcomes.

For these data, the multivariate model added little to the significance tests. Because the residual correlation between turnout and Democratic share is nearly zero ($r = 0.05$), the joint tests gave almost the same answers as the separate regressions, changing the conclusion only for the weak female-share effect. That is itself an answer to the project's question: when the outcomes' unexplained parts are

essentially unrelated, a multivariate model offers little benefit for testing which predictors matter. The joint model was still valuable for the cross-outcome tests, testing whether a predictor's effects on turnout and share are equal or not.

The results have several limitations. The ACS data are 2020–2024 estimated averages with sampling error measured at the district level rather than for individual voters. For example, the finding that districts with more college graduates have higher turnout does not show that college graduates themselves are more likely to vote, only that there is a correlation between the number of college graduates and voter turnout. This period also includes the pandemic, which may have affected the averages. The model also omits key variables, such as prior partisanship and district unemployment rates, which could improve its explanatory power. A final limitation is the districts in the same state are likely to have related errors since they share the same election rules which can result in clustered errors, resulting in standard errors and p-values that are too small.

Future research could add measures the model lacks, such as prior partisan vote and unemployment rates, adjust the Pillai tests for within-state correlation (for example, by resampling whole states in a bootstrap), or repeat the analysis for other election years to see whether the patterns hold.

In the 2024 House elections, district characteristics explained most of the variation in both turnout (71%) and Democratic share (82%). Education, race and ethnicity, age, density, inequality and incumbency were all jointly significant. Most predictors affected the two outcomes by different amounts or in opposite directions, separating the drivers of participation from those of partisanship. Because the outcomes' residuals were nearly uncorrelated, the multivariate and separate-regression approaches reached the same conclusions for all but one weak effect; the multivariate model's main contribution was the ability to test cross-outcome differences directly.

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Appendix A: Variable Definitions

CVAP is the citizen voting-age population. The last three variables were excluded from the model because of collinearity with median income.

Variable	Label	Definition
turnout	Turnout (%)	House votes cast as a share of CVAP
dshare	Democratic two-party share (%)	Democratic votes / (Democratic + Republican votes)
pct_black	Black share (%)	Share of CVAP that is Black
pct_hisp	Hispanic share (%)	Share of CVAP that is Hispanic
pct_asian	Asian share (%)	Share of CVAP that is Asian
pct_ba	Bachelor's degree or higher (%)	Share of adults with a bachelor's degree or higher
inc_10k	Median household income	Median household income, in $10,000s
gini	Income inequality (Gini index × 100)	Gini index of household income, times 100
pct_18_29	Age 18–29 (%)	Share of CVAP aged 18–29
pct_65plus	Age 65 and over (%)	Share of CVAP aged 65 and older
pct_female	Female share (%)	Share of CVAP that is female
log_density	Population density (log10)	Log base 10 of persons per square mile
inc_dem	Democratic incumbent	1 if a Democratic incumbent ran (open seat = baseline)
inc_rep	Republican incumbent	1 if a Republican incumbent ran (open seat = baseline)
pct_poverty	Poverty rate (%) (excluded)	Share of people below the poverty line
pct_low_inc	Low-income households (%) (excluded)	Share of households with low income
pct_high_inc	High-income households (%) (excluded)	Share of households with high income

Appendix B: Diagnostic and Supplementary Figures

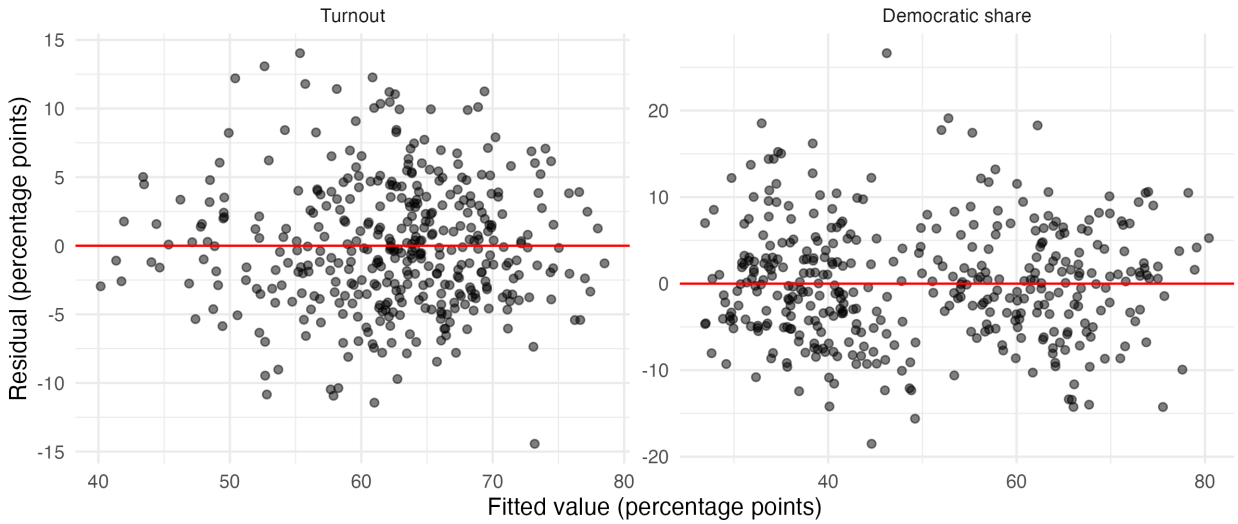


Figure 1: Residuals vs. fitted values for turnout and Democratic share

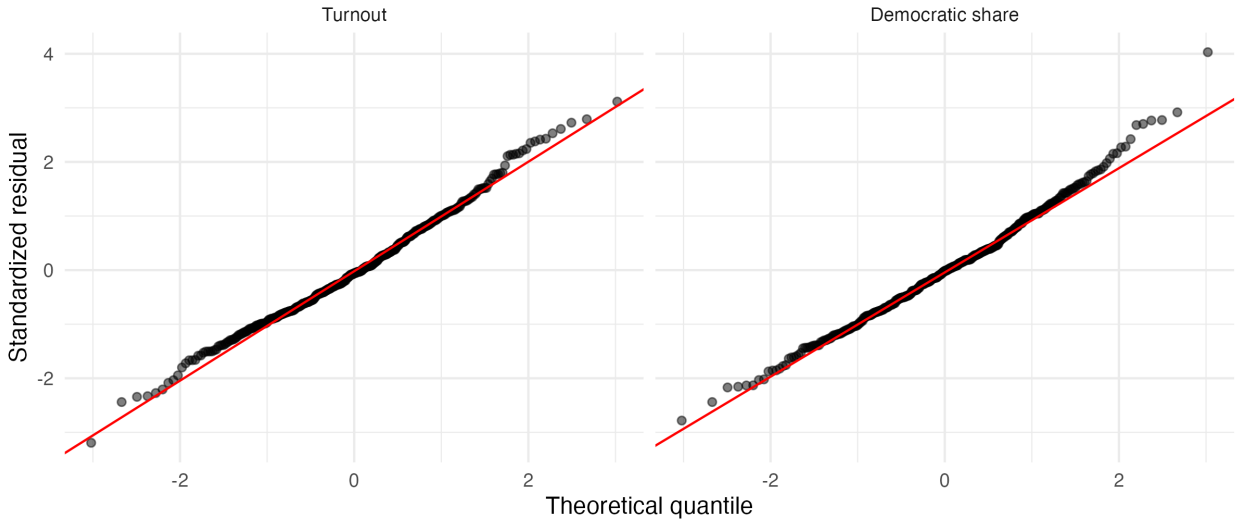


Figure 2: Normal Q-Q plots of standardized residuals for turnout and Democratic share

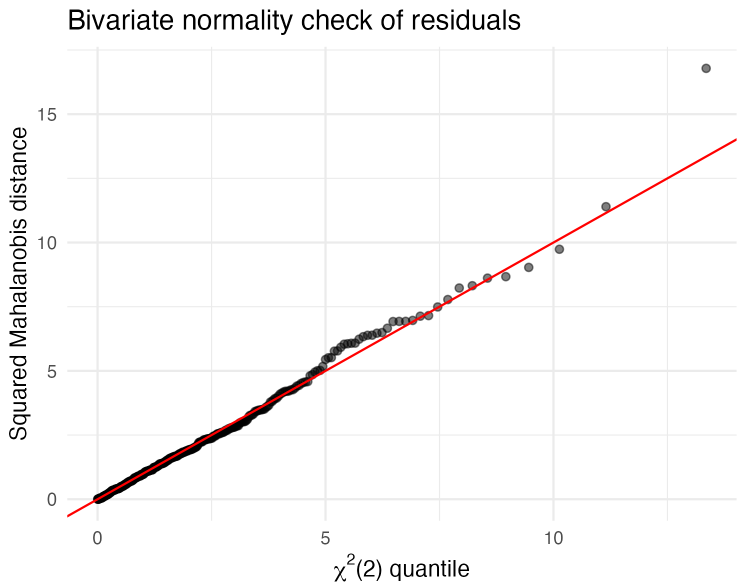


Figure 3: Bivariate normality check: squared Mahalanobis distances of the residual pairs vs. $^2(2)$ quantiles